



NAKUJA PROJECT

Solid Propulsion Team

KNSB Grain Preparation Criteria

Surfactant-Modified Rheology with Vacuum Degassing

Document status: Revision A — Proposed, pending experimental verification
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Supersedes: Grain Prep Procedure, April 5, 2026

Purpose

This document defines the step-by-step preparation procedure for one KNSB composite sugar propellant grain and the acceptance criteria at each stage. It is the working reference for the Solid Propulsion Team during grain curing sessions. Two process modifications are introduced in this revision: surfactant addition and vacuum degassing of the cast grain. Both are adopted from literature and the Nakuja vacuum curing report, and are subject to the comparative verification programme described in Section 4.

Ingredient and Equipment Checklist

Before starting, verify every item below is present and in acceptable condition. Do not proceed if any line cannot be checked.

Reagents

Ingredient	Required mass	Acceptance criterion
Sorbitol	690.87 g	Dry, no discolouration
Potassium nitrate (fine)	944.84 g	Fine powder; grind at low setting if coarse
Surfactant (SLS or equiv.)	0.87 g	Single-ingredient, known purity; not a consumer blend

Surfactant dose check: the maximum permitted dose is **0.053 g per 100 g of dry mix**. For this batch ($m_{\text{dry}} = 1635.71 \text{ g}$): $m_{\text{surfactant}} = 1635.71 \times 0.053/100 \approx 0.87 \text{ g}$. Do not exceed this value. The dose of 0.158 g/100 g caused self-extinguishment in the EUCASS study.



Equipment

Item	Check before use
Skillet pan and induction stove	Clean; stove calibration not overdue
Electrical balance	Zeroed; resolution ≤ 0.01 g
Silicone spatula	Clean; no residue from prior batches
Casting mold, liner, core rod	Assembled; release grease applied
Stopwatch	Functioning
Vacuum pump (VE115N)	Nameplate confirmed as VE115N; oil level checked
Vacuum chamber (21 L pot)	Acrylic lid, gasket, and all fittings free of cracks or damage
Vacuum gauge (liquid-filled)	Readable; not cracked
Ball valves $\times 2$	Both seats and handles functional
PTFE tape	Applied to all threaded fittings
PPE (gloves, eye protection)	Worn by all personnel
Fire extinguisher	Present and accessible

Preparation Procedure and Stage Criteria

Each stage lists the required actions and the indicators of a good or poor outcome. A **FAIL** indicator means stop and investigate before continuing.

Stage 1 — Sorbitol Melt

Actions:

1. Place the skillet pan on the induction stove.
2. Weigh 690.87 g of sorbitol into the pan. Set the stove to 150 °C.
3. Heat for 30 minutes, then reduce to 120 °C and begin stirring.
4. Watch for the colour change to caramel (5 minutes at 120 °C; allow 1–2 minutes extra if needed).

Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Melt colour at 120 °C	Uniform light caramel	Pale/unmelted areas or dark brown (FAIL)
Melt consistency	Smooth, low viscosity	Lumps or crystalline patches (FAIL)
Time to caramel	5–7 min	>10 min or instant darkening (CAUTION)
Odour	Mild caramel	Sharp burning smell (FAIL — overheated)



Stage 2 — Surfactant Addition

Actions:

1. Once caramelisation is confirmed, reduce temperature to 90 °C.
2. Add the pre-weighed surfactant (0.87 g) to the molten sorbitol.
3. Stir continuously for approximately 1 minute until fully dispersed.

Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Surfactant incorporation	Fully dissolved, no separate phase	Visible undissolved film on surface (CAUTION)
Viscosity after addition	Noticeable decrease (flows more freely)	No change in viscosity (CAUTION)
Colour change	None / minimal	Darkening or precipitation (FAIL)

Stage 3 — KNO₃ Incorporation

Actions:

1. Begin adding 944.84 g of fine KNO₃ in small increments while stirring continuously.
2. After 1 minute of powder addition, increase temperature back to 120 °C.
3. Continue stirring. After all KNO₃ has been added and 20 minutes have elapsed, reduce to 60 °C while stirring.

Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Mixture homogeneity	Uniform grey-tan, no white pockets	Visible white KNO ₃ lumps (FAIL)
Pourability at 60 °C	Flows smoothly off spatula	Stiff, pulls or tears (CAUTION)
Mixture colour	Consistent throughout	Streaking or dark patches (CAUTION)
Reaction signs	None	Sparkling, popping, smoke (FAIL — stop immediately)

Stage 4 — Casting

Actions:

1. Transfer the mixture into the assembled casting mold and insert the core rod.
2. Do **not** fully fasten the top plate — leave a vent gap so gas can escape into the surroundings before and during vacuum degassing.



Indicator	Good (PASS)	Poor (FAIL / CAUTION)
Fill level	Mold filled to top; no voids visible at surface	Air pockets or incomplete fill (CAUTION)
Surface finish	Smooth, flat top surface	Rough, pitted, or sunken (CAUTION)
Vent gap	Present; top plate loose or offset	Top plate fully sealed (FAIL — remove before degassing)
Time from mix to casting	≤ 5 min	> 10 min (grain may have stiffened — CAUTION)

Stage 5 — Vacuum Degassing

Actions:

1. Within 30–60 minutes of casting (while the grain is still warm and pliable), place the vented mold inside the vacuum chamber.
2. Close and seat the acrylic lid on its gasket. Confirm both ball valves are closed.
3. Open the isolation valve and start the pump. **Draw down gradually** — a fast pull risks foaming.
4. Target working vacuum: **10–50 mbar**. Do not pull toward the pump's ≈ 0.2 mbar ultimate vacuum.
5. Hold and observe through the lid. Dwell target: **10–15 minutes**, or until bubbling clearly stops.
6. Close the isolation valve. Slowly crack open the vent valve to return to atmospheric pressure gradually.
7. Remove the mold and fully fasten the top plate.



Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Achieved vacuum	10–50 mbar	>50 mbar (insufficient draw — CAUTION); <5 mbar (excessive — CAUTION)
Pump-down behaviour	Steady gauge drop; slight bubbling at surface	Rapid foaming reaching lid (FAIL — vent immediately)
Bubbling at dwell end	None or minimal	Vigorous, sustained bubbling (CAUTION — extend dwell)
Vent-back behaviour	Gradual pressure rise, no shock	Sudden pop or implosion sound (FAIL — inspect lid and gasket)
Lid and gasket post-use	No cracks, no deformation	Any crack or failure (FAIL — take out of service)

Stage 6 — Final Cure

Actions:

1. Set the sealed, degassed grain aside in a dry, well-ventilated storage area.
2. Cure for **5–7 days** before any inspection or use.

Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Cure duration	5–7 days	<5 days (FAIL — do not use)
Storage conditions	Dry, shaded, no ignition sources nearby	Humid, exposed to sun or heat (FAIL)
Mold integrity at end of cure	No cracks, no seepage	Propellant seepage at seams (FAIL)

Post-Cure Acceptance Criteria

Density and Porosity Check

Compute the theoretical void-free density from ingredient mass fractions:

$$\frac{1}{\rho_{\text{theoretical}}} = \frac{w_{\text{KNO}_3}}{\rho_{\text{KNO}_3}} + \frac{w_{\text{sorbitol}}}{\rho_{\text{sorbitol}}}$$

For this batch: $w_{\text{KNO}_3} = 0.578$, $w_{\text{sorbitol}} = 0.422$, giving $\rho_{\text{theoretical}} \approx 1.81 \text{ g/cm}^3$.

Measure each grain's mass and geometric volume (do not use water immersion — sorbitol is moisture-sensitive). Express the result as a percentage of $\rho_{\text{theoretical}}$.



Criterion	Target	Outcome
Density ratio $\rho_{\text{actual}}/\rho_{\text{theor}}$	$\geq 97\%$	$< 95\% = \text{FAIL}$
Improvement vs. control batch	Higher % = better	Worse than control = CAUTION

Visual Inspection (Cross-Section)

Section at least one grain per batch and inspect visually.

Indicator	Good (PASS)	Poor (FAIL/CAUTION)
Internal voids	None visible	Any large void $> 2\text{ mm}$ (FAIL); small pitting (CAUTION)
Microcracks	None	Any crack reaching the bore surface (FAIL)
Bore surface	Smooth, circular, concentric	Rough, eccentric, or chipped (FAIL)
Colour uniformity	Uniform throughout cross-section	Banding, inclusions, or dark streaks (CAUTION)

Verification Test Matrix

Before adopting this revised procedure for production, validate it empirically across four conditions with at least $n = 3$ grains per condition.

Run	Condition	Purpose
A	Baseline (no surfactant, no vacuum)	Control
B	Surfactant only (0.053 g/100 g)	Isolate rheology/wetting effect
C	Vacuum only (no surfactant)	Isolate degassing effect
D	Surfactant + vacuum (this revision)	Combined effect

Compare density ratios and cross-section quality across all four runs. Adopt condition D for production only if it demonstrates a statistically meaningful improvement over the baseline in both the density check and the visual inspection.

Safety Requirements

All of the following apply at all times during grain preparation. Non-compliance is grounds for stopping the session immediately.

- Full PPE (gloves, eye protection) worn by all personnel at all times.
- No open flame, smoking, or unnecessary ignition sources in the work area.



- Fire extinguisher present and accessible throughout.
- Work in a well-lit, ventilated area; no distractions (phones, music).
- KNO_3 ground at low grinder setting only; never at high speed (sparking risk).
- Pure ingredients only; no substitution of consumer blends.
- Vacuum pump motor and electrics kept as far from propellant and KNO_3 dust as the tubing run allows.
- Mold vent gap confirmed small enough that no powder or molten propellant can be drawn into the vacuum tubing.
- Pressure changes (draw-down and vent-back) performed gradually in all cases.
- Acrylic lid, gasket, and fittings inspected before each session; do not operate a cracked or damaged chamber.
- First batch under this revised procedure must be a reduced-scale test grain, not a full production grain.

Sign-Off

Sign-off	Prepared by: _____
	Reviewed by: _____
	Approved by: _____
	Date: _____
	Revision: _____